

AI that uses renewable energy and/or least energy usage

Posted by u/v0lcanine • 0 points • 12 comments

So I have been incredibly anti-AI for years, especially for art. But due to me pursuing a career in video game art and many studios use it and I dont want to be left behind, I have to cave in. I am looking for a model that uses renewable energy or the least amount of energy for the results. I will probably be using it to make concept art for my 3D models or, if available, do my UV mapping, or other shortcuts for efficiency without taking my craft away from me. What models should I use? Any recs?

Comments

u/aiArt-ModTeam • 1 points • 2026-01-05 16:01 UTC

A comprehensive list of AI engines is readily available in the welcome thread, group sidebar, and within every post. <https://docs.google.com/spreadsheets/d/1zYJUM-srhglA7wrj4Pe4QqepAsHIEC00DydoTPv4PWg/> Repeated requests for recommendations create unnecessary clutter. Members who repeatedly ignore this rule may face a permanent ban.

u/Aadi_880 • 5 points • 2026-01-05 03:31 UTC

If that's your concern, I'm pretty sure your best bet is *running an AI model yourself.*

You don't need to use corporate LLMs from the USA that run on massive data centers. AI models exist that can run ***on your laptop.*** Environmental impact is pretty much non-existent. That being said, it is recommended to have above average specs to do this. You already need a fairly decent PC to make graphics heavy games to begin with.

These models are made by the open source community. There's tons of them, mostly using and iterating over Chinese models and gets very competitive results.

I say that because it's really hard to quantify environmental impact of, well, anything. Since doing anything (including traditional art) has some impact. I chose this way of local generation because any impacts is essentially meaningless in the grand-scheme of things.

Z-Image-Turbo is the current hype in effecient models right now for consumer grade systems. However, folks at the **StableDiffusion** subreddit can help you better than here.

u/v0lcanine • 1 points • 2026-01-05 03:57 UTC

Thank you so much!

u/_Luminous_Dark • 3 points • 2026-01-05 03:28 UTC

You can run AI image/video/text generation models locally on your own computer. It'll use the same energy source as the rest of your house, and you can make that renewable.

u/v0lcanine • 1 points • 2026-01-05 03:55 UTC

Ok! Another poster gave me some sources to learn how. Next question would be to ask which models to use?

u/KallyWally • 1 points • 2026-01-05 04:40 UTC

Depends on what you need. I don't know of any model that can do UV mapping specifically, but I found [this texturing tool](<https://github.com/FrederikHasecke/diffused-texture-addon>) which might be interesting.

There also models that can make... uh, models. AI models that can make 3D models, you get it. But from what I've heard the topology isn't great.

Concept art is probably the easiest. Sixhaunt gave you a good rundown of the current top models, I'll add Chroma and Illustrious/NoobAI (or your preferred finetune) to the list if you want something more 2D style.

u/Sixhaunt • 2 points • 2026-01-05 03:26 UTC

The ones you would be using for commercial purposes are local models you run on your computer so if you run your computer on renewable energy then every model is using renewable energy for you.

The state of the art models right now use about as much electricity as playing a modern video game for the same time so if you are using it for work then I don't see any reason it would be worse than using the same electricity to play a game.

I would suggest you look into setting up ComfyUI on your computer and try some various models locally on your computer. If your computer isn't too good then look into quantized versions of the models (they are minified versions that run on lower end software without too much loss in quality).

the r/StableDiffusion subreddit is a good place to ask about setting this kind of stuff up and it's 100% free and running on your computer without internet so if you don't like AI, you won't be giving a single cent to an AI company and you control if it's running on renewables or not based on if your household is running on renewables.

u/v0lcanine • 2 points • 2026-01-05 03:53 UTC

This is a fantastic answer thank you so much. I have never thought about or known I could run it locally. I def would be interested in setting it up as to not use those data centers and give support to companies.

u/Sixhaunt • 1 points • 2026-01-05 04:16 UTC

Glad I could help!

If you want some suggestions for free open source models to try once you get it setup, the z-image-turbo model is great and so are the Qwen-image ones. Z-Image is my favourite for text2image right now, but there's a good qwen-image-edit model meant for making edits to existing images which isn't as easy with pure text2image models like z-image. My sister is graphic designer and has been loving the edit models for prototyping and stuff so she can test out designs or get some ideas that she puts into

the final product.

the edit models allow you to supply 1 or more images as inputs and when you write a prompt you can reference the images to ask for elements from multiple images combined together or take one of the images and just have it change certain aspects, etc...

The text2image models on the other hand are just turning prompts into images and so the only way to really use them for editing an image is if you do "inpainting" which is like photoshop's Content-aware fill where you select an area and have it generate only that one section while making it fit with the surrounding areas. Unlike content-aware-fill, you can control what it gets filled with using a prompt rather than it automatically fetching parts of the current image to splice into the region.

Generally I find the text2image models like z-image do better at generating an image from nothing and follows the prompt really well and allows for very long multi-paragraph prompts with a ton of detail. The image edit models can generate from scratch but generally I find it to not be quite as good for that but you can take an image you make with z-image then use an edit model to make specific changes until you get precisely what you want.

I also think I should point out that when you first get started with the common models and tools like comfyUI, it may look a bit intimidating and here's an example of a portion of a workflow for inpainting:

[https://preview.redd.it/7820scothgbg1.png?
width=2130&format=png&auto=webp&s=bcd07b1941d4fb89c1c2f90fcaad1eb1f9860e1](https://preview.redd.it/7820scothgbg1.png?width=2130&format=png&auto=webp&s=bcd07b1941d4fb89c1c2f90fcaad1eb1f9860e1)

Although it looks overwhelming (and trust me, workflows can be WAY larger and more complex than this), you can get very far by just changing a few options like the prompt and image size. You don't need to learn about step counts, CFG, schedulers, samplers, shift values, Loras, controlnets, etc... until you're comfortable with the models.

u/Electrical_Face_1737 • 2 points • 2026-01-05 03:21 UTC

I saw some lady doing a tiktok on her work on something like this, basically a raspberrypi with solar doing some yard work, when communication doesn't have to be instant like chat you can have slower models that take a minute to predict what you would like it to do or say or call an api etc.

u/jpwne • 1 points • 2026-01-05 08:47 UTC

If you call an api you are just offloading to an energy source you have no say over. The only way to have full energy control over usage is to run locally. Life cycle usage is down to what hardware you choose to buy etc.

u/AutoModerator • 1 points • 2026-01-05 03:12 UTC

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https://www.reddit.com/r/aivideos/comments/1kfhxf/ regarding_the_other_ai_video_group/

* Looking for an AI Engine? Check out our MEGA list [here]

(<https://docs.google.com/spreadsheets/d/1zYJUM-srhgIA7wrj4Pe4QqepAsHIEC00DydoTPv4PWg/>)

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Hope everyone is having a great day, be kind, be creative!

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